

# Harindu Bandara

Altimerra | +94777255476 | harindudb@gmail.com | harindu-bandara  
0009-0009-3360-8980 | BSc | Sri Lanka



*Mechanical engineer seeking opportunities to further expand my knowledge and skills, interested in pursuing a career in R&D related to the fields of Robotics and Bioengineering.*

## Education

 **Bachelor of Science in Mechanical Engineering** 2020 - 2024  
UNIVERSITY OF MORATUWA Katubedda

- GPA: 3.52
- Dean's List: Semester 8
- Research: Development of a Trans-radial prosthetic device
- Major Coursework: Robotics | Biomedical Engineering | Control Systems | Computer-Aided Engineering
- Extra-curricular: Karate | Classical Music Society

 **CIMA Certificate Level** 2019  
WISDOM BUSINESS ACADEMY Colombo

- Financial Accounting | Management Accounting | Economics | Business Law

 **G.C.E. Advanced Level** 2016 - 2018  
ANANDA COLLEGE Colombo

- Member of: Science Society | IT Society

 **G.C.E. Ordinary Level** 2004 - 2015  
VIDURA COLLEGE Sri Jayawardenepura Kotte

- Extra-curricular: Scouting

## Professional Experience

 **Temporary Instructor** 2025  
UNIVERSITY OF MORATUWA Katubedda

- Department of Mechanical Engineering

Biomedical Engineering   Robotics   Design of Machine Elements   Advanced Manufacturing  
Mechanics of Machines   Manufacturing Engineering

 **Visiting Instructor** 2024  
UNIVERSITY OF MORATUWA Katubedda

- Faculty of Engineering
- Faculty of Medicine

Control Systems   ECG   Op Amps

 **Design Engineer (Intern)** 2023  
ARIMAC LANKA Kollupitiya

- Worked in a team that designed the navigation and intelligence of the Diyazen robot

ROS   Python   C++   AI

## Publications

- [1] G. M. H. D. Bandara, D. D. Amarasinghe, H. A. N. P. Hettiarachchi, H. H. M. J. D. Silva, R. K. P. S. Ranaweera, and R. A. R. C. Gopura, "Dexterous Robotic Finger with Joint-Mobility Modulation for Adaptive Grasping," in *Moratuwa Engineering Research Conference (MERCon)*, 2024.

## Conferences

 **MERCon** 2024  
UNIVERSITY OF MORATUWA Katubedda

- Presenting Author: Dexterous Robotic Finger with Joint-Mobility Modulation for Adaptive Grasping
- Best paper nomination for Mechanical Systems track

## Projects and Associations

- **Design of a Trans-radial Prosthetic Device**  
BIONICS LAB, UNIVERSITY OF MORATUWA
- 2023 - 2024  
Katubedda
- Development of a patent-pending mechanism
  - Extensive research on background and technical aspects
  - Detailed design, simulations and mathematical modelling
  - Prototype fabrication and real-world testing
  - Control system design and implementation using C++ and Python
- **Industrial Optimization Project**  
LANKEM PLC
- 2024  
Ja-Ela
- Field visits, data gathering
  - Analysis, optimization and process simulations
- **Development of a Heart-on-a-chip MEMS device**  
DEPARTMENT OF MECHANICAL ENGINEERING
- 2023  
Katubedda
- Design of a Microfluidic Chip and a Piezoelectric diaphragm
  - Flow analysis and structural analysis using COMSOL
- **Design of a gearbox for a Jeweler's Rolling Mill**  
DEPARTMENT OF MECHANICAL ENGINEERING
- 2022  
Katubedda
- 3D modelling of gearbox
  - Programmatic calculation of design parameters using Julia

## Skills and Certificates

English Cambridge English | IELTS  
Research Rapid Research | Knowledge Management  
Analysis Finite Element | Dynamic Systems | Control Systems  
Design 3D Modelling | Mechanical Design | Control System Design  
Writing Research papers | Patents | Technical reports  
AI and ML Data preprocessing | Training and Model Selection | Model Application  
Prog. Lang. C++ | Python | Julia | MATLAB | JavaScript

## Scholarships and Awards

- **University Scholarship Scheme**  
CEYLON PETROLEUM STORAGE TERMINALS LTD.
- 2021 - 2024
- **Price for Academic Performance**  
ANANDA COLLEGE
- 2019
- **Prize for best performance in Mathematics and English**  
VIDURA COLLEGE
- 2015

## References

- **Prof. Ruwan Gopura**  
UNIVERSITY OF MORATUWA
- Dean, Faculty of Graduate Studies
  - gopurar@uom.lk
- **Prof. Ranjith Amarasinghe**  
UNIVERSITY OF MORATUWA
- Head of Department, Department of Mechanical Engineering
  - ranama@uom.lk
- **Dr. Pubudu Ranaweera**  
UNIVERSITY OF MORATUWA
- Senior Lecturer, Department of Mechanical Engineering
  - pubudur@uom.lk